

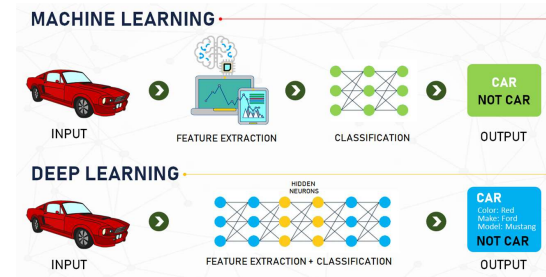


Convolutional Neural Networks

shailesh s

shaileshsivan@gmail.com

Deep learning



27-May-26

Convolutional Neural Networks

2

Why Deep Learning useful?

- Manually designed features are often **over-specified**, **incomplete** and take a **long time to design** and validate
- Learned Features are **easy to adapt**, **fast** to learn
- Deep learning provides a very **flexible**, (almost?) **universal**, learnable framework for representing world, visual and linguistic information.
- Effective **end-to-end** joint system learning
- Utilize large amounts of training data

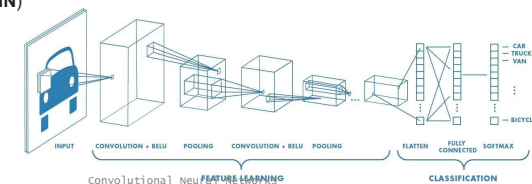
27-May-26

Convolutional Neural Networks

3

Convolutional Neural Networks

- A convolutional neural network (CNN, or ConvNet) is a class of artificial neural network, most commonly applied to analyze visual imagery
- They are also known as **shift invariant** or **space invariant artificial neural networks (SIANN)**



27-May-26

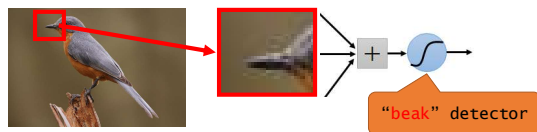
Convolutional Neural Networks

4

Consider learning an image:

- Some patterns are much smaller than the whole image

Can represent a small region with fewer parameters



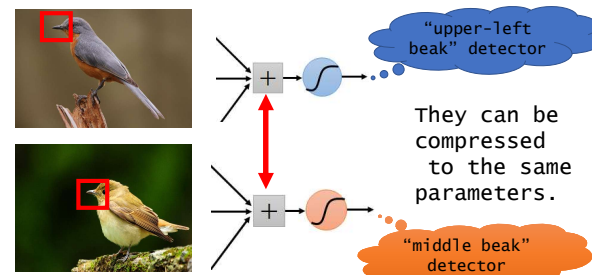
27-May-26

Convolutional Neural Networks

5

Same pattern appears in different places:
They can be compressed!

What about training a lot of such "small" detectors and each detector must "move around".



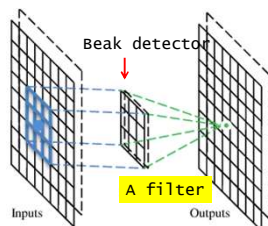
27-May-26

Convolutional Neural Networks

6

A convolutional layer

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that does convolutional operation.



27-May-26

Convolutional Neural Networks

7

Convolution

These are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Filter 2

Each filter detects a small pattern (3 x 3).

27-May-26

Convolutional Neural Networks

8

Convolution

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Dot product



Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image



1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

Convolution

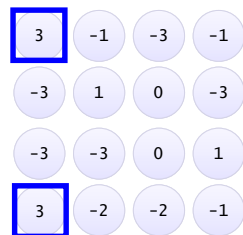
stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



Convolution

stride=1

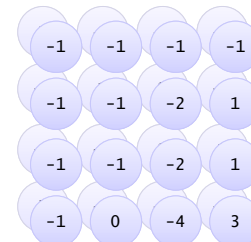
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

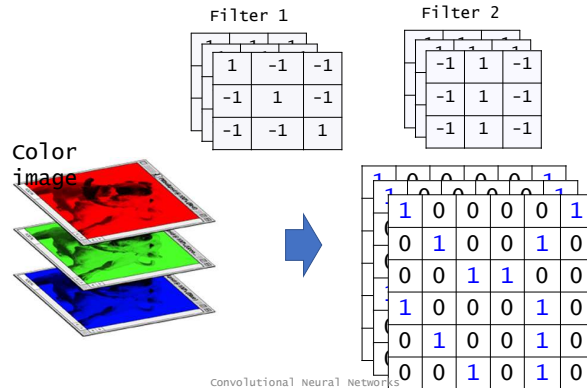
Repeat this for each filter



Two 4 x 4 images
Forming 2 x 4 x 4
matrix

Feature Map

color image: RGB 3 channels

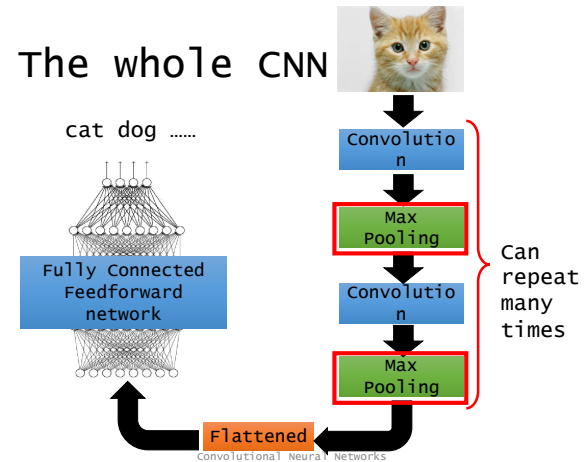


27-May-26

Convolutional Neural Networks

13

The whole CNN

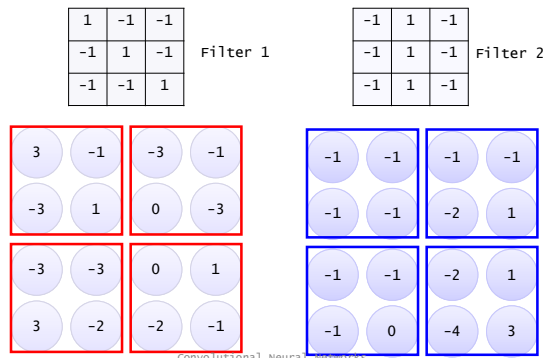


27-May-26

Convolutional Neural Networks

14

Max Pooling



27-May-26

Convolutional Neural Networks

15

why Pooling

- Subsampling pixels will not change the object



we can subsample the pixels to make image smaller

→ fewer parameters to characterize the image

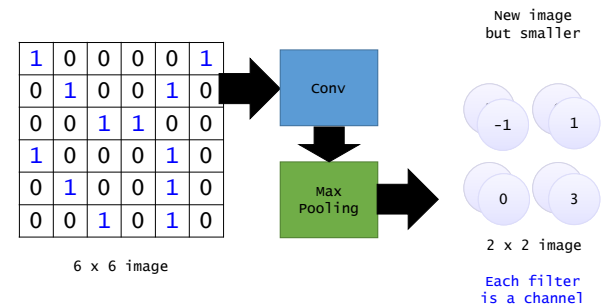
27-May-26

Convolutional Neural Networks

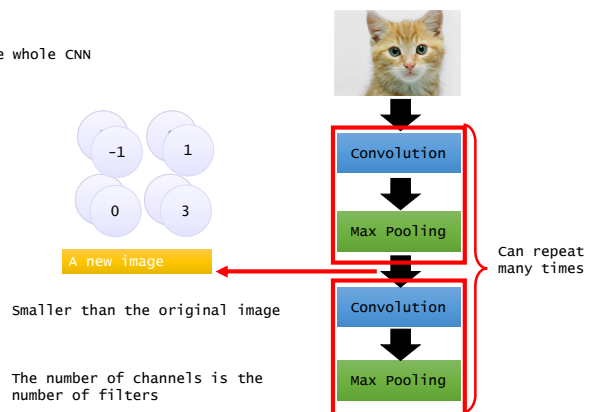
A CNN compresses a fully connected network in two ways:

- Reducing number of connections
- Shared weights on the edges
- Max pooling further reduces the complexity

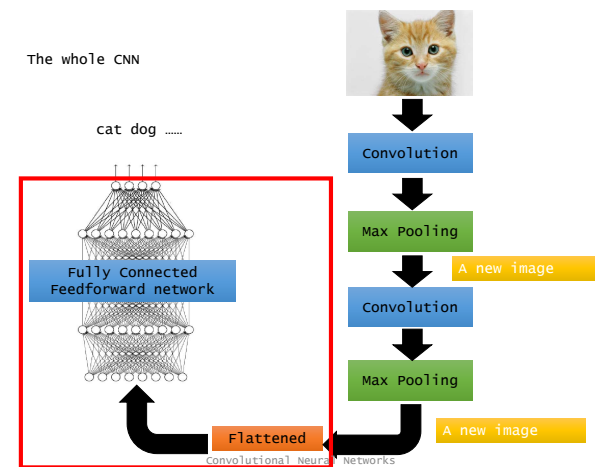
Max Pooling

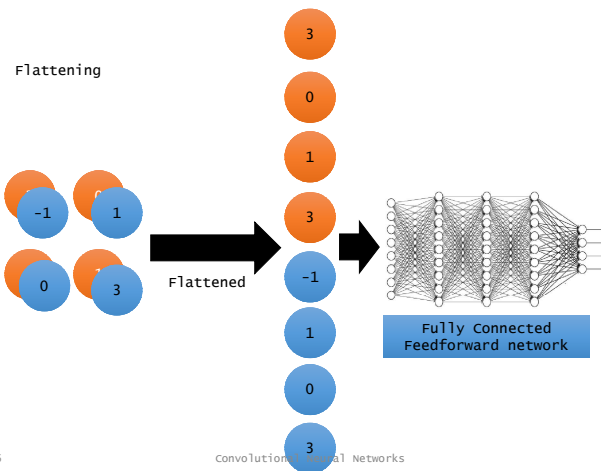


The whole CNN



The whole CNN



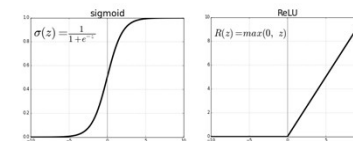


27-May-26

Convolutional Neural Networks

21

ReLU activation



- Rectified Linear Unit
- Computationally efficient
- Non-linear
- The sigmoid and hyperbolic tangent activation functions cannot be used in networks with many layers due to the vanishing gradient problem
- The rectified linear activation function overcomes the vanishing gradient problem, allowing models to learn faster and perform better

27-May-26

Convolutional Neural Networks

22

Thank You



27-May-26

Convolutional Neural Networks

23